

An Ontological Proof of Becoming

constructed with no axioms

Primitive. A *law* is a map m on a carrier. Write $\text{Fix}m$ for “ m , read as an equation, has a solution” ($\exists x. x = m x$), and $\text{Run}m$ for “ m , read as a recursion, has a solution from every seed and none of them is stationary” (for all x the sequence $x, mx, m(mx), \dots$ is infinite and no term equals the next).

Df. 1. m is *structureless* iff m is definable from a bare two-valued distinction alone, naming neither value: equivalently, m commutes with the exchange of the two values.

Df. 2. A tokening is *absolute* iff its mark admits no outside: no second locus, at the one time of the tokening, for its contrast to be realized in.

Df. 3. m is *the ground* iff m is structureless and non-trivial (not the identity).

Ax. 1 (Arena). A carrier admits a non-trivial structureless law only if it has exactly two values. *[one value marks nothing; three or more collapse every relabeling-invariant law to the identity]*

Ax. 2 (Uniqueness). On two values the only structureless laws are the identity and the negation $\neg$; the identity is satisfied by everything and so asserts nothing. Hence the ground, if it exists, is $\neg$.

Ax. 3 (Retorsion). The tokening of “no distinction is drawn,” as against its own denial, itself draws one. The premise of Df. 2 cannot be coherently refused — its negation is unsayable, not false.

Ax. 4 (No margin). An absolute tokening (Df. 2) has no outside at its own moment; if its contrast is realized at all, it is realized at another time.

Th. 1 (No being). $\neg \text{Fix}(\neg)$. *[the two values are distinct and each fails its own negation; no state is identical to its own negation]*

Th. 2 (Becoming, uniquely). $\text{Run}(\neg)$, and the orbit from every seed is the same period-two alternation, unique up to which value opens it. *[write down $x, \neg x, x, \neg x, \dots$; uniqueness is induction on the step]*

Th. 3 (Equivalence). For the ground, $\neg \text{Fix}(\neg) \leftrightarrow \text{Run}(\neg)$. *[a fixed point is a one-state orbit that never moves; a stalled orbit is a fixed point: refusing to be and being unable to stop running are one hypothesis, read in two frames]*

Th. 4 (Actuality). Let a tokening deny that $\neg$ runs. By Ax. 3 it is tokened as against its own denial, hence is a tokening in the sense of Df. 2, hence absolute; by Ax. 4 its contrast is realized at another time; by Ax. 1–2 the only realizable contrast on two values is $\neg$ itself. So the tokened denial of Th. 2 is, at two times, an instance of Th. 2.

Corollary. Necessarily, a becoming exists; by Th. 1 no being answers to the ground concept; by Th. 3 these are one theorem; and by Th. 4 the becoming cannot be coherently denied without the denial itself being one of its ticks.

All of the above is a named theorem or definition of a Lean 4 development that is axiom-free and available at [GitHub](#).